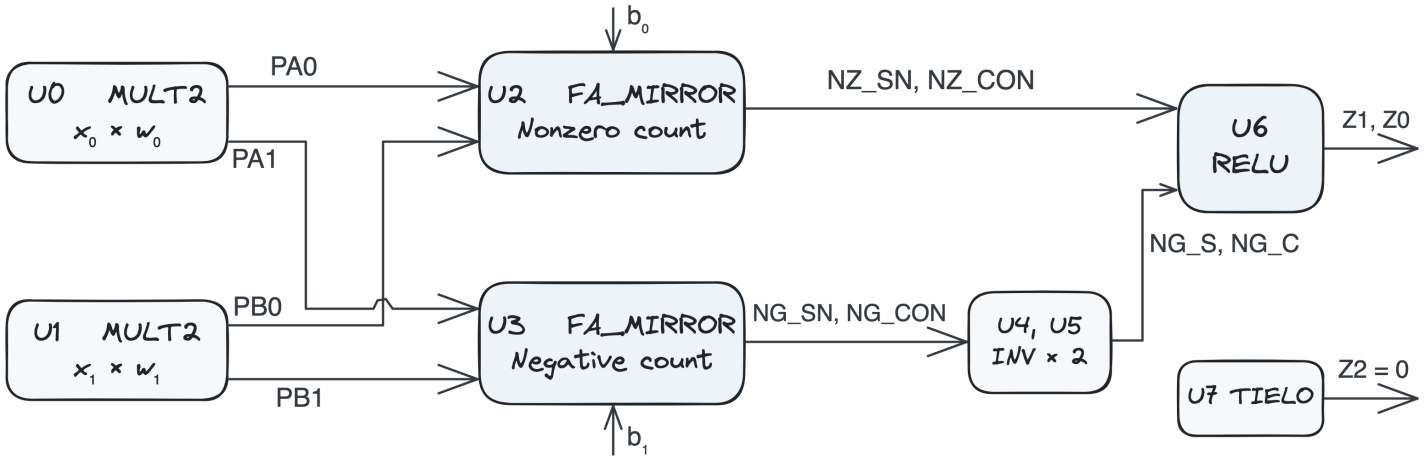
Milestone 1 Gate-Level Neuron Design

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ECE 555 Group 20

Neuron schematic

The neuron computes $Z = \max(0, x_0w_0 + x_1w_1 + b)$. Each input, weight and bias uses the two-bit code 11 for −1, 00 for 0, or 01 for +1. For each product, bit 0 tells us whether the product is nonzero and bit 1 tells us whether it is negative.

Carry-save datapath



U2 counts the nonzero terms and U3 counts the negative terms, including the bias. A negative term contributes once to each count. Subtracting twice the negative count from the nonzero count gives the signed sum. The two adders can count in parallel.

Transistor count

Cell	Instances	Transistors per cell	Total
MULT2	2	22	44
FA_MIRROR	2	24	48
INV	2	2	4
RELU	1	14	14
TIELO	1	2	2
Neuron			112

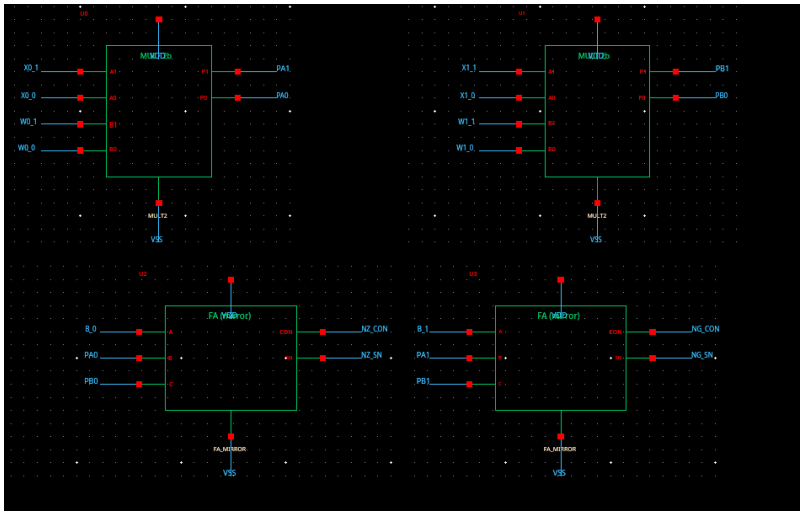
Device sizing

All cells use ASAP7 RVT devices with a 20 nm gate length. Each NMOS has 2 fins (54 nm) and each PMOS has 3 fins (81 nm). PMOS bodies connect to VDD and NMOS bodies to VSS.

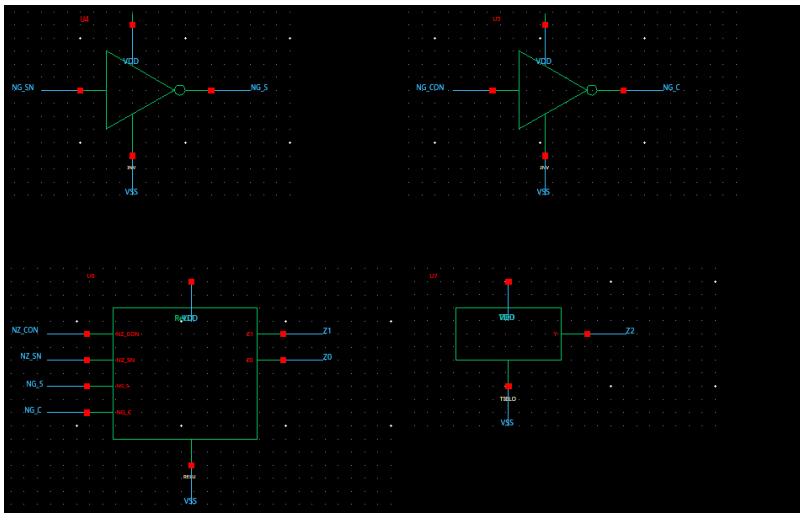
Top-level NEURON schematic

U0 and U1 produce PA and PB. U2 takes the low bits of PA, PB and the bias. U3 takes the high bits. Both mirror adders have inverted sum and carry outputs. U4 and U5 restore the polarity of the negative count before it reaches RELU.

Multipliers and full adders (U0–U3)



Output logic (U4–U7)



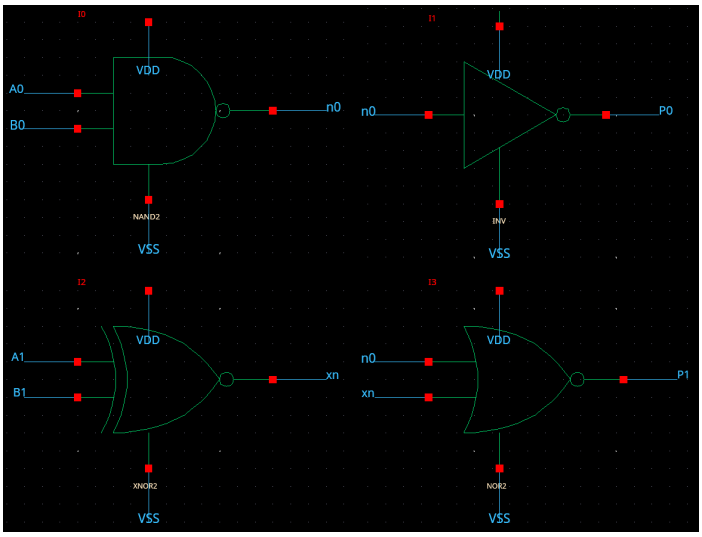
Instance connections

Inst.	Cell	Input connections	Output connections
U0	MULT2	A1=X0_1, A0=X0_0, B1=W0_1, B0=W0_0	P1=PA1, P0=PA0
U1	MULT2	A1=X1_1, A0=X1_0, B1=W1_1, B0=W1_0	P1=PB1, P0=PB0
U2	FA_MIRROR	A=B_0, B=PA0, C=PB0	CON=NZ_CON, SN=NZ_SN
U3	FA_MIRROR	A=B_1, B=PA1, C=PB1	CON=NG_CON, SN=NG_SN
U4	INV	A=NG_SN	Y=NG_S
U5	INV	A=NG_CON	Y=NG_C
U6	RELU	NZ_CON=NZ_CON, NZ_SN=NZ_SN, NG_S=NG_S, NG_C=NG_C	Z1=Z1, Z0=Z0
U7	TIELO		Y=Z2

Matching net names in the schematics indicate a connection. The largest sum is 3 (011), so U7 holds Z2 low. U6 drives Z1 and Z0.

Two-bit multiplier

MULT2 schematic



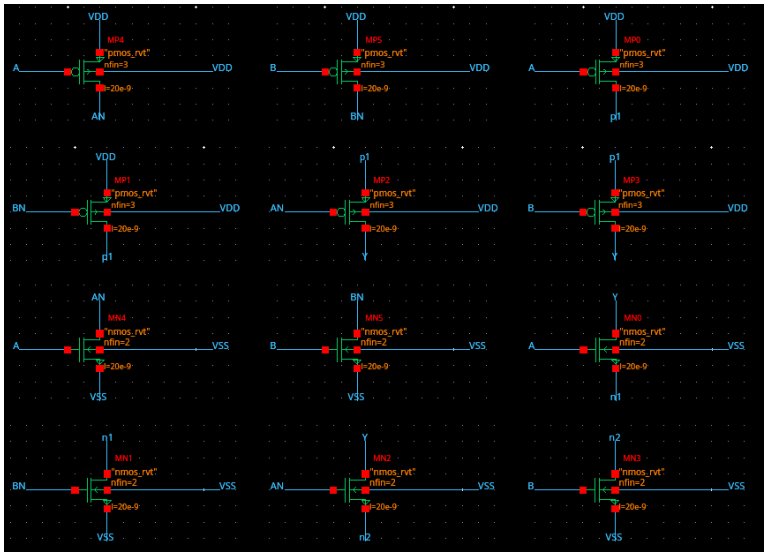
P0 is high when both operands are nonzero. P1 is high when P0 is high and the signs differ. I0 and I1 form P0. I2 compares the sign bits, and I3 combines its output with the NAND output to form P1.

Inst.	Cell	Input connections	Output connections
I0	NAND2	A=A0, B=B0	Y=n0
I1	INV	A=n0	Y=P0
I2	XNOR2	A=A1, B=B1	Y=xn
I3	NOR2	A=n0, B=xn	Y=P1

Product truth table

A \ B	11 (−1)	00 (0)	01 (+1)
11 (−1)	01	00	11
00 (0)	00	00	00
01 (+1)	11	00	01

XNOR2 schematic

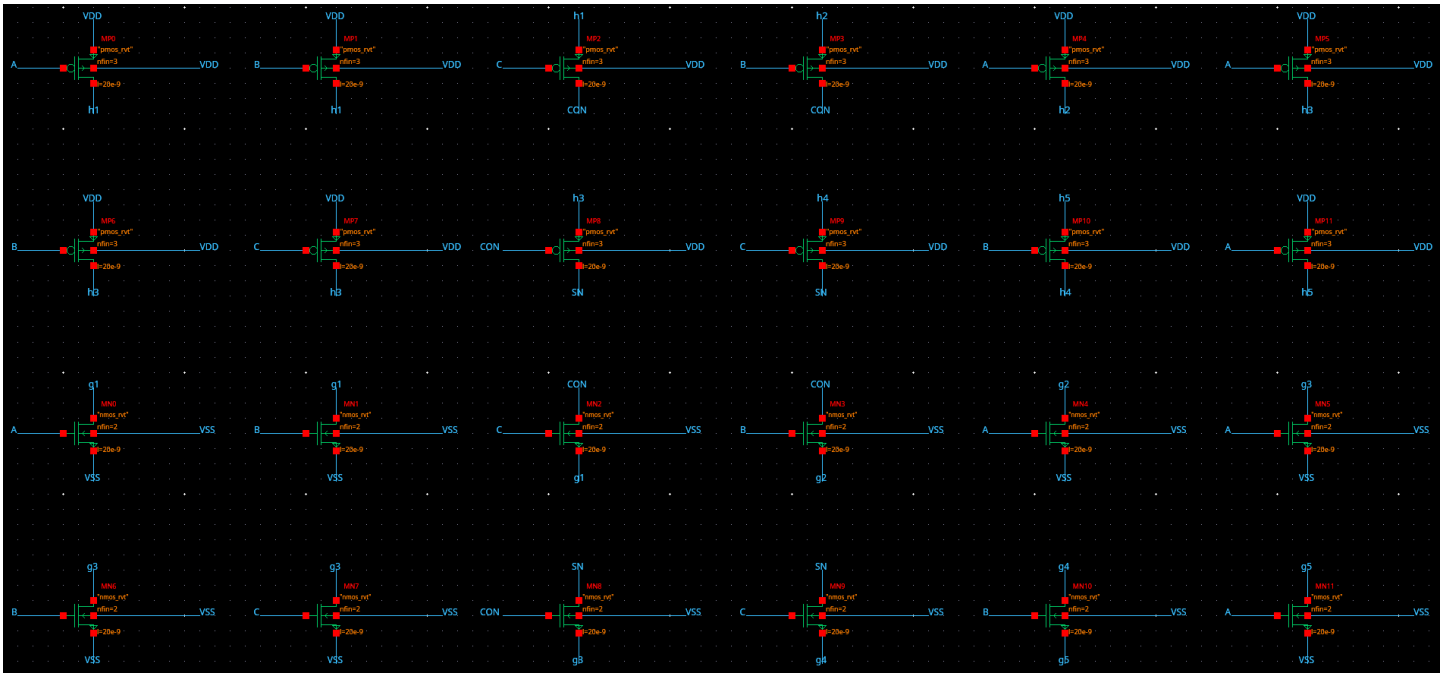


XNOR2 uses two input inverters and an eight-transistor AOI22 core. The output is high for equal signs. The NOR2 in MULT2 then forces P1 low.

Mirror full adder

FA_MIRROR uses 24 transistors. CON goes low when at least two inputs are high. The sum network uses CON and the three inputs to produce SN, the complement of the sum bit.

FA_MIRROR schematic



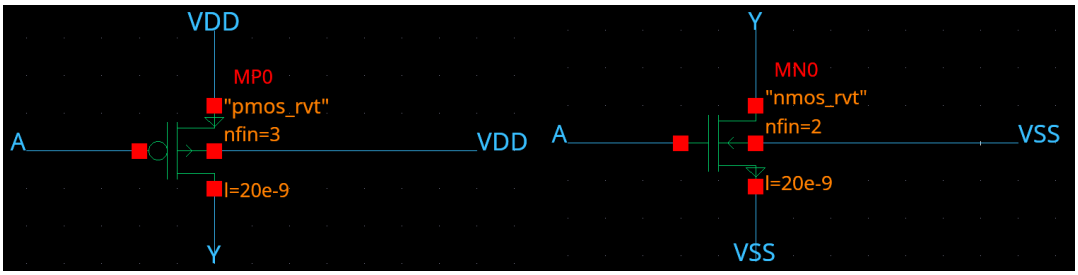
Transistor connections

CON drives MP8 and MN8 in the sum network. Each PMOS/NMOS pair in the table shares its gate net.

PMOS / NMOS	Gate	PMOS drain	PMOS source	NMOS drain	NMOS source
MP0 / MN0	A	h1	VDD	g1	VSS
MP1 / MN1	B	h1	VDD	g1	VSS
MP2 / MN2	C	CON	h1	CON	g1
MP3 / MN3	B	CON	h2	CON	g2
MP4 / MN4	A	h2	VDD	g2	VSS
MP5 / MN5	A	h3	VDD	g3	VSS
MP6 / MN6	B	h3	VDD	g3	VSS
MP7 / MN7	C	h3	VDD	g3	VSS
MP8 / MN8	CON	SN	h3	SN	g3
MP9 / MN9	C	SN	h4	SN	g4
MP10 / MN10	B	h4	h5	g4	g5
MP11 / MN11	A	h5	VDD	g5	VSS

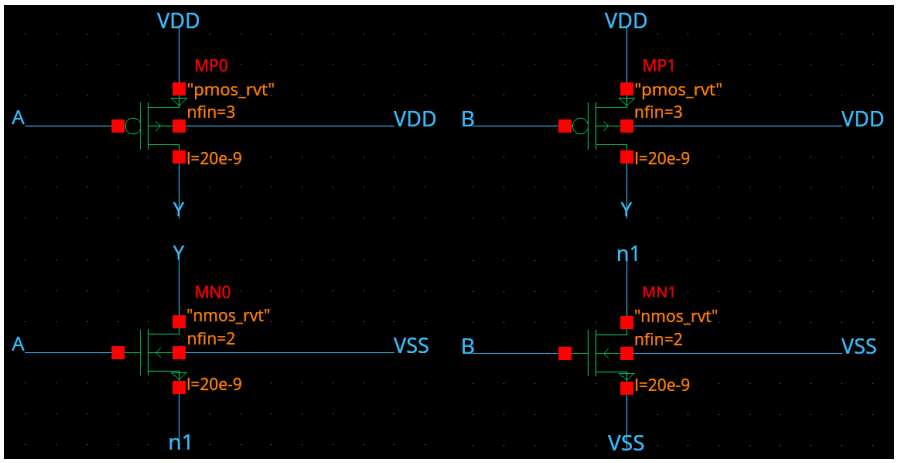
Inverter and two-input gates

INV schematic



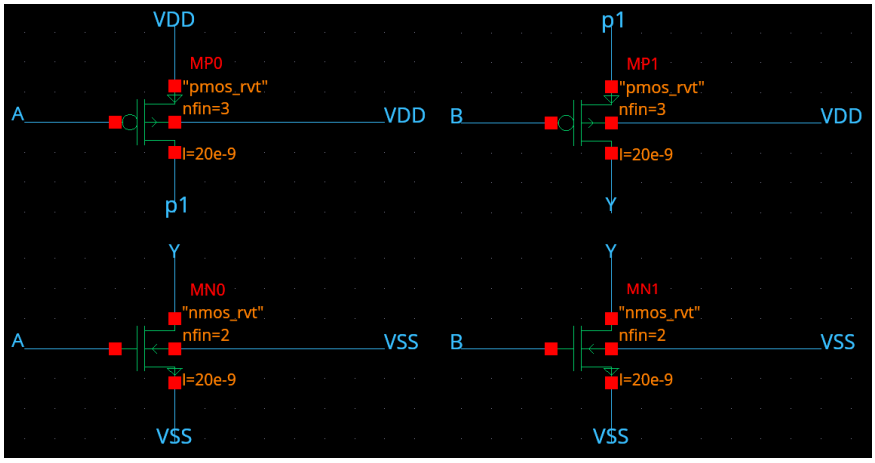
Both gates connect to A and both drains connect to Y. The PMOS pulls Y high for A = 0. The NMOS pulls Y low for A = 1.

NAND2 schematic



The NMOS devices meet at n1 and pull Y low only when A and B are both high. The two PMOS devices connect in parallel from VDD to Y.

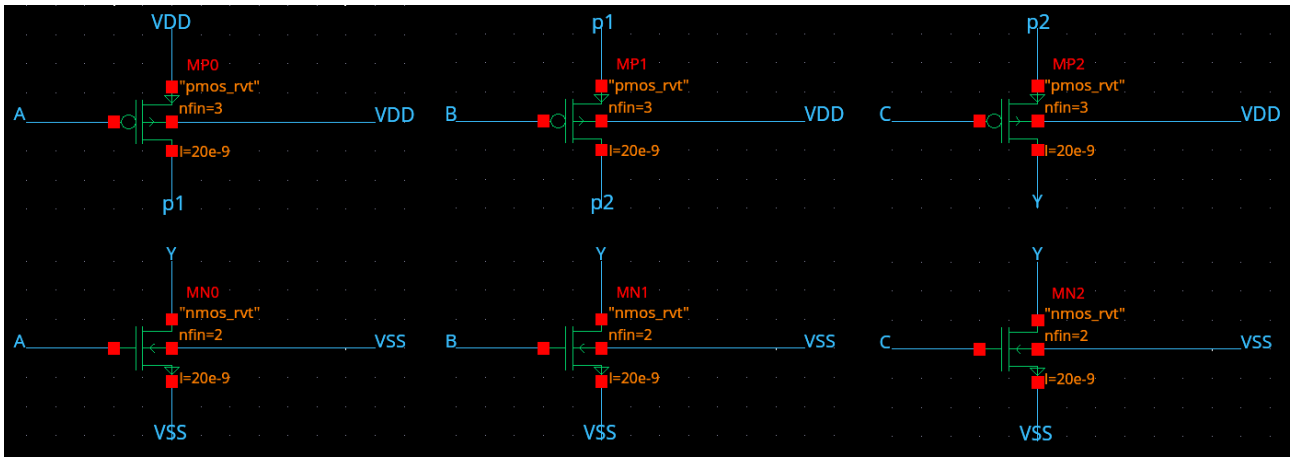
NOR2 schematic



The PMOS devices meet at p1 and pull Y high only when both inputs are low. Either NMOS can discharge Y to VSS.

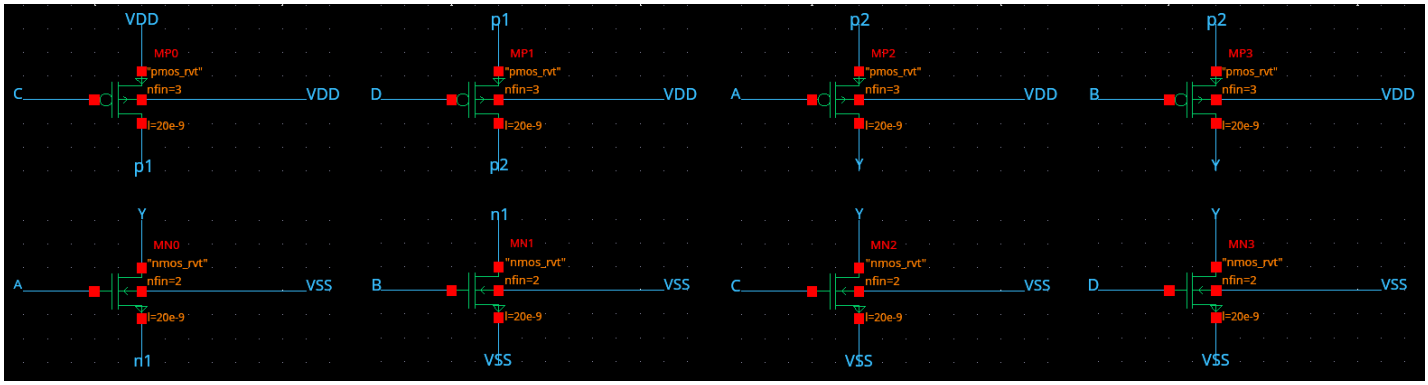
ReLU gates and tie-low cell

NOR3 schematic



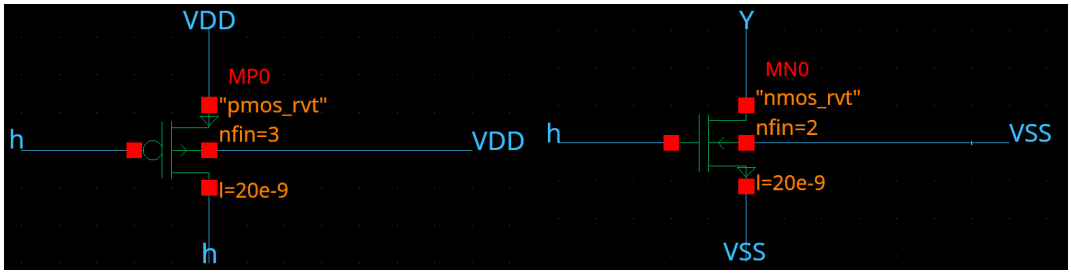
Three series PMOS devices pull Z1 high when NZ_CON, NG_S and NG_C are all low. Any high input turns on one of the parallel NMOS devices.

AOI211 schematic



Z0 goes low when NZ_CON and NG_S are both high, or when either NZ_SN or NG_C is high. The first condition uses the series NMOS pair. Each remaining input drives a separate NMOS path to VSS.

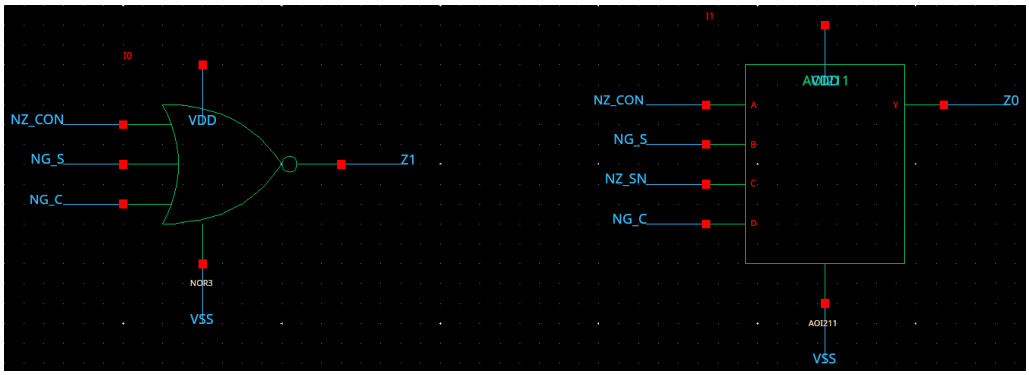
TIELO schematic



The diode-connected PMOS biases node h high. The NMOS gate connects to h and pulls its drain, Z2 at the top level, to VSS.

ReLU output logic

RELU schematic



I0 drives Z1 high if at least two terms are nonzero and none are negative. I1 drives Z0 high for sums of 1 and 3. The inverted outputs from the nonzero counter feed these gates directly.

Inst.	Cell	Input connections	Output connections
I0	NOR3	A=NZ_CON, B=NG_S, C=NG_C	Y=Z1
I1	AOI211	A=NZ_CON, B=NG_S, C=NZ_SN, D=NG_C	Y=Z0

ReLU truth table

Nonzero terms	Negative terms	Signed sum	Z2 Z1 Z0
0	0	0	000
1	0	1	001
1	1	-1	000
2	0	2	010
2	1	0	000
2	2	-2	000
3	0	3	011
3	1	1	001
3	2	-1	000
3	3	-3	000

Functional verification

The switch-level check and the pre-layout HSPICE simulation each gave the expected output for all 243 input combinations (three values for each of x_0, w_0, x_1, w_1 and b). The HSPICE testbench used the TT models at 0.9 V and 25 °C with a 1 fF load on each output. It held each input combination for 400 ps and sampled 5 ps before the next input change.

Across all three outputs, logic lows reached at most 33 µV and logic highs measured at least 0.900 V at the sample time.